

# Aiden Slabiak

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## EDUCATION

**University of Alabama**

December 2026

*B.S. Computer Science*

*GPA: 3.7/4.0*

**Forsyth Central High School (Dual Enrollment at University of North Georgia)**

June 2023

*GPA: 4.1/4.0*

## COURSEWORK

**Courses:** Complex Algorithms & Data Structures, Operating Systems & Concurrency, High-Performance Computing, Programming Languages (Compiler Design)

**Awards:** Dean's List (3x)

## TECHNICAL SKILLS

**Languages:** C/C++, Python

**Tools:** Git, GitHub, Bitbucket, Jira, Confluence, VS Code, GDB, WSL, Salesforce

**Libraries/Technologies:** OpenMP, MPI, Boost.Asio, NumPy, pandas, Matplotlib

## PROJECTS

**6-DOF Flight Dynamics Simulator** | *C++, Python*

Mar. 2026

- Implemented 6-degree-of-freedom rigid-body flight simulation for rockets and missiles in C++.
- Modeled aerodynamic drag, thrust, gravity, and coordinate frame transformations.
- Integrated modular propulsion and aerodynamic systems with visualization capabilities.

**SIMD Accelerated Matrix Library** | *C++, AVX2/AVX-512, Intrinsics*

Jan. 2026

- Vectorized 4x4 and 8x8 matrix operations using AVX2 compiler intrinsics.
- Achieved ~3.5x speedup over scalar loops via loop unrolling and 32-byte alignment.
- Tested extensively with random matrices to ensure numerical correctness.

**Aviation Delay Analytics Pipeline** | *Python, Streamlit, Data Engineering*

Apr. 2026

- Built a multi-source aviation analytics pipeline to process and model historical flight delay datasets.
- Developed a Streamlit dashboard for real-time visualization of delay trends and operational metrics.
- Optimized large-scale data processing workflows to reduce latency and improve predictive performance.

## EXPERIENCE

**Undergraduate Research Assistant** | *University of Alabama*

2026 – Present

- Contributed to the development of an AI-driven platform for students.
- Researched and evaluated modern AI techniques to inform platform design decisions.
- Collaborated with faculty and graduate researchers to validate findings and ensure accuracy of insights.

**Software Developer Intern** | *Scientific Research Corporation*

Summer 2026

- Engaged in full-lifecycle development within the Salesforce ecosystem, encompassing data modeling, process automation, and user interface customization.
- Built and optimized complex business logic using **Salesforce Flow** to automate internal business processes.
- Collaborated with senior developers to evaluate system configurations and logic tradeoffs, gaining exposure to enterprise-grade software deployment and testing.

## ADDITIONAL ACHIEVEMENTS

**Study Abroad – East & Southeast Asia**

Summer 2025

- Conducted a market analysis between developing economies in Vietnam and established markets in Taiwan
- Engaged with international business leaders and university faculty in Tokyo and Taipei to evaluate the integration of emerging technologies within diverse regulatory and economic environments.